

Deepak Kushwaha

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SUMMARY

Senior Software Development Engineer with 10+ years building Amazon's internal big data development platform for analytics and business data production. Lead platform services that let data engineers schedule, orchestrate, monitor, and recover Redshift and EMR workloads at scale, supporting **50K+ engineers** and **5M+ workflow executions/day**.

EXPERIENCE

Big Data Technologies, Amazon

Mar. 2023 – Present

Senior Software Development Engineer

Seattle, WA

- **AI-Powered Operations Platform:** Led an Amazon Bedrock RAG platform for **30K+ developers**, automating root-cause analysis across **3M+ ETL executions/day** and reducing support by **50+ tickets/week**.
- **Workflow Orchestration Platform:** Led redesign of Big Data Technologies' data pipeline and job orchestration platform, which schedules and executes **5M+ analytics jobs/day** for **50K+ Amazon data builders**; architected the scalable engine using an internal workflow service, EventBridge, DynamoDB, Kinesis, and queues for scheduling, retries, distributed state, and fault recovery.
- Improved platform reliability from **99.0% to 99.99%** while leading a **5–6 engineer** cross-functional initiative, delivering a highly observable and extensible orchestration framework.

Big Data Technologies, Amazon

Jul. 2019 – Mar. 2023

Software Development Engineer II

Seattle, WA

- **Dependency Management & Data Completeness Platform:** Led redesign of Big Data Technologies' dependency platform powering **5M+ analytics workflow executions/day** with fine-grained dataset completeness and declarative execution dependencies.
- Built a lineage and completeness platform aggregating **50M+ completeness events/day**, with optimized storage for low-latency dependency evaluation, blocking-dataset detection, workflow recovery, automated redrive, and end-to-end debugging.

Amazon

Jul. 2016 – Jul. 2019

Software Development Engineer I / II

Bengaluru, India

- **Amazon Appstore Ingestion Platform:** Designed backend services for validating, processing, transcoding, and publishing Android application packages (APKs) submitted by external developers.
- Built distributed ingestion workflows for secure onboarding and release pipelines, improving reliability, throughput, and developer experience.

EDUCATION

National Institute of Technology Karnataka (NITK), Surathkal

2012 – 2016

Bachelor of Technology in Computer Science & Engineering

Surathkal, India

SKILLS

Languages : Java, Python, Kotlin, TypeScript

Distributed Systems & Cloud : Workflow Orchestration, Event-Driven Architecture, DynamoDB, Kinesis, EventBridge, Lambda, ECS, S3, Docker, Kubernetes

AI & Data Platforms : LLMs, Agentic AI, RAG, Amazon Bedrock, Apache Spark, Apache Iceberg, Redshift